Prob. 5. (2&T. Ch 2, §5.1)

Prove that the set of f-ns $\cos(nx)$, $\sin(nx)$ (for $n = 1, 2, 3, \ldots$) & constants are orthogonal over the interval $[-\pi, \pi]$ by confirming the ↓ 8 integral identities by hand:

(a) $\int_{-\pi}^{\pi} \cos(nx) \cdot 1 \, dx = 0$ (This shows that $\cos(nx)$ & const.s are $\perp$)

Sol-n: $I = \int_{-\pi}^{\pi} \cos(nx) \cdot 1 \, dx = \int_{-\pi}^{\pi} \cos(nx) \, dx = \left[\frac{\sin(nx)}{n} \right]_{-\pi}^{\pi} =$

$= \frac{1}{n} \left[\underbrace{\sin(n\pi)}_{0} - \underbrace{\sin(-n\pi)}_{0} \right] = \frac{1}{n} \cdot 0 \equiv 0$ ✓ $\Rightarrow$

$\langle \cos(nx), 1 \rangle = 0.$

(b) $I = \int_{-\pi}^{\pi} \sin(nx) \cdot 1 \, dx = \int_{-\pi}^{\pi} \sin(nx) \, dx = \left[-\frac{\cos(nx)}{n} \right]_{-\pi}^{\pi} =$

$= -\frac{1}{n} \left[\cos(n\pi) - \overset{\text{even f-n}}{\cos(-n\pi)} \right] = -\frac{1}{n} \left[\cos(n\pi) - \cos(n\pi) \right] \equiv 0$ ✓

$\Rightarrow \langle \sin(nx), 1 \rangle = 0.$

(c) $I = \int_{-\pi}^{\pi} \cos(nx) \cdot \sin(kx) \, dx$; $k \neq n$, & $k = n$.

① Let $k \neq n$, & $k, n \in \mathbb{N}$

$I = \int_{-\pi}^{\pi} \cos(nx) \sin(kx) \, dx = \int_{-\pi}^{\pi} \frac{1}{2} \left[\sin((k-n)x) + \sin((k+n)x) \right] dx =$

$= \frac{1}{2} \left\{ \int_{-\pi}^{\pi} \sin[(k-n)x] \, dx + \int_{-\pi}^{\pi} \sin[(k+n)x] \, dx \right\} =$

$\cos(-x) = \cos(x)$

$$= \frac{1}{2}\left\{ \left[-\frac{\cos[(k-n)x]}{k-n} \right]_{x=-\pi}^{\pi} + \left[-\frac{\cos[(k+n)x]}{k+n} \right]_{x=-\pi}^{x=\pi} \right\} \qquad \underline{\underline{k \neq n \Rightarrow}} \quad k-n \neq 0$$

$$= \frac{1}{2}\left\{ 0 + 0 \right\} \equiv 0 \quad \Rightarrow$$

(1) $\langle \sin(kx), \cos(nx) \rangle = 0$ when $n \neq k$.

(2) Now let $k = n$, then

$$I = \int_{-\pi}^{\pi} \sin(kx) \cdot \cos(nx)\, dx = \int_{-\pi}^{\pi} \sin(nx) \cdot \cos(nx)\, dx = \frac{1}{2}\int_{-\pi}^{\pi} \sin(2nx)\, dx$$

$$= \frac{1}{2}\left[\frac{\cos(2nx)}{-2n} \right]_{x=-\pi}^{x=\pi} = \frac{1}{-4n}\left[\cos(2\pi n) - \cos(-2\pi n) \right] = \frac{-1}{4n} \cdot 0 \equiv 0, \Rightarrow$$

(2) $\langle \sin(kx), \cos(nx) \rangle = 0$ when $n = k$.

Thus by (1) & (2), we have $\langle \sin(kx), \cos(nx) \rangle \equiv 0 \quad \forall k, n.$

(d) $I = \int_{-\pi}^{\pi} \cos(nx) \cdot \cos(kx)\, dx, \quad k \neq n.$ $\longrightarrow$ $\boxed{k-n \neq 0}$ so

$$I = \int_{-\pi}^{\pi} \frac{1}{2}\left[\cos[(n-k)x] + \cos[(n+k)x] \right] dx = \frac{1}{2}\int_{-\pi}^{\pi} \cos[(n-k)x]\, dx +$$

$$+ \frac{1}{2}\int_{-\pi}^{\pi} \cos[(n+k)x]\, dx = \frac{1}{2}\left[\frac{\sin[(n-k)x]}{n-k} \right]_{x=-\pi}^{x=\pi} + \frac{1}{2} \cdot \left[\frac{\sin[(n+k)x]}{(n+k)} \right]_{x=-\pi}^{x=\pi} =$$

$\boxed{k, n \in \mathbb{N} \text{ so } \sin(n \pm k)\pi = 0}$

$$= \frac{1}{2} \cdot [0] + \frac{1}{2} \cdot [0] = 0. \quad \Rightarrow$$

$$I = \langle \cos(nx), \cos(kx) \rangle = 0, \quad n \neq k.$$

for $n \neq k$ & $n, k \in \mathbb{N}$

(e) $I = \int_{-\pi}^{\pi} \sin(nx)\cdot\sin(kx)\,dx = \int_{-\pi}^{\pi} \frac{1}{2}\left\{\cos[(n-k)x] - \cos[(n+k)x]\right\} dx =$

$= \frac{1}{2}\int_{-\pi}^{\pi} \cos[(n-k)x]\,dx - \frac{1}{2}\int_{-\pi}^{\pi} \cos[(n+k)x]\,dx \; (=)$ ← $\boxed{n, k \in \mathbb{N},\ k \neq n \Rightarrow k \pm n \in \mathbb{Z}}$

$(=) \frac{1}{2}\left[\frac{-\sin[(n-k)x]}{(n-k)}\right]_{x=-\pi}^{\pi} - \frac{1}{2}\left[\frac{-\sin[(n+k)x]}{(n+k)}\right]_{x=-\pi}^{x=\pi} =$ ← $\left[\begin{array}{l}\sin(\pm 5\pi) = 0 \\ \Rightarrow \sin(\pm n\pi) = 0\end{array}\right]$

$= \frac{1}{2}\cdot 0 - \frac{1}{2}\cdot 0 = 0 \Rightarrow$

$I = \langle \sin(nx), \sin(kx) \rangle = 0$ when $n \neq k$. $x \in [-\pi, \pi]$

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(f) $I = \int_{-\pi}^{\pi} \cos^2(nx)\,dx = \int_{-\pi}^{\pi}\left[\frac{1}{2} + \frac{1}{2}\cdot\cos(2nx)\right]dx = \left[\frac{x}{2}\right]_{-\pi}^{\pi} + \frac{1}{2}\cdot\left[\frac{\sin(2nx)}{2n}\right]_{-\pi}^{\pi}$

$= \frac{1}{2}\left[\pi - (-\pi)\right] + \frac{1}{2}\cdot[0] = \pi. \Rightarrow$

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$I = \langle \cos(nx), \cos(nx) \rangle = \pi$ when $n \in \mathbb{N}$.

(g) $I = \int_{-\pi}^{\pi} \sin^2(nx)\,dx = \int_{-\pi}^{\pi} \frac{1}{2} - \frac{1}{2}\cos(2nx)\,dx = \left[\frac{x}{2}\right]_{-\pi}^{\pi} - \frac{1}{2}\cdot\left[\frac{\sin(2nx)}{2n}\right]_{x=-\pi}^{x=\pi} =$

$= \frac{1}{2}\left[\pi - (-\pi)\right] - \frac{1}{2}\cdot[0] = \pi \Rightarrow$

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$I = \langle \sin(nx), \sin(nx) \rangle = \pi$ when $n \in \mathbb{N}$ & $x \in [-\pi, \pi]$.

(h) $I = \int_{-\pi}^{\pi} 1\,dx = [x]_{x=-\pi}^{x=\pi} = \pi - (-\pi) = 2\pi \Rightarrow$ ✓

$I = \langle 1, 1 \rangle = 2\pi$ when $x \in [-\pi, \pi]$, as desired.

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